SPECIFICATION

Background of the Invention

Forward secrecy is a critical property of secure communication protocols, ensuring that the compromise of a long-term secret key does not compromise past session keys. This is typically achieved through key exchange protocols like Diffie-Hellman, which generate ephemeral session keys. In the new paradigm of physical layer security using disposable encoding maps, a similar mechanism is needed. If a single disposable map is used for a very long session, an attacker could potentially capture enough data to perform frequency analysis or other attacks. A protocol is needed to limit the lifetime of any single encoding map to ensure forward secrecy at the encoding layer.

Summary of the Invention

The present invention is a protocol for encoding map rotation that provides forward secrecy for communication systems using disposable encoding maps on high-cardinality channels. The protocol ensures that any single encoding map is used for only a limited period or for a limited amount of data, thwarting cryptanalytic attacks that require large samples of data encoded with the same map.

The method operates as an enhancement to a secure communication session that uses a disposable encoding map derived from a shared secret. The protocol defines a trigger condition for map rotation. This trigger can be a certain number of data frames transmitted, a specific amount of time elapsed, or a combination of both. The communicating endpoints continuously monitor for this trigger.

When the trigger condition is met, both endpoints, which share a secret seed and a deterministic algorithm, can independently and synchronously generate the *next* encoding map in a pre-determined sequence. They then seamlessly switch from the old map to the new map at a designated frame number or time. The old map is then securely discarded. This process repeats throughout the communication session. An attacker who compromises the shared secret at time T would only be able to generate the map for the current epoch (Map N), but would not be able to reconstruct the previous, discarded maps (Map N-1, Map N-2, etc.), and thus cannot decode previously intercepted traffic.

Brief Description of the Drawings

FIG. 1 is a diagram illustrating the process of encoding map rotation during a communication session.

Detailed Description of the Invention

Referring to **FIG. 1**, the protocol is shown in operation over time. A communication session (101) begins between two endpoints using a first disposable map, Map A (102), which is generated from a shared secret. The endpoints transmit a series of data frames (103) encoded with Map A. A trigger condition (104) is defined, for example, a count of 10,000 frames. As the session proceeds, the endpoints count the transmitted frames. When frame 10,000 is reached, the trigger is activated. Both endpoints independently generate the next map in the sequence, Map B (105), and discard Map A. All subsequent data frames (106) are transmitted using Map B, until the next trigger condition is met, at which point they would switch to Map C, and so on. This periodic rotation ensures that any single map is used for only a limited window of communication.